# AutoVet: Real-World Problems Q&A

Practice questions a panel may ask: a real clinic problem, and how AutoVet handles it. Every answer describes what the system actually does today.

> **How to answer:** name the problem, say what AutoVet does, and give one concrete detail (a number, a rule, or where it shows on screen). If there is a limit, say it plainly; panels trust honest answers more than perfect ones.

## 1. Internet and server problems

**Q: The clinic's internet goes down in the middle of the day. Can staff keep working?**

**A:** Yes, on the clinic computer. It runs AutoVet on its own local database, and the admin app opens at `localhost/clinic` with no internet. Staff can register clients, book, record visits and bill. Every change is queued, and when the internet returns, `php artisan app:sync` (run every minute by the scheduler) sends the changes to the cloud and brings down any changes made online.

> **Be honest:** Sync is built and tested, but on the live server it is switched off until it is configured in Render. It supports one clinic computer.

**Q: A client books online through the portal while the clinic computer is offline. Is the booking lost?**

**A:** No. The booking is saved in the cloud. When the clinic computer reconnects, the sync pulls it down, linked to the right pet and owner, and it appears in the clinic's appointment list.

**Q: The same client's address is edited at the clinic and online before they sync. Whose edit is kept?**

**A:** The latest edit wins, on both sides. Each record carries the time it was last changed, and the newer version replaces the older one, so the clinic and the cloud always end up agreeing.

**Q: The clinic computer and the cloud both create invoice VB-2026-11-0005 while disconnected. What happens?**

**A:** Both invoices are kept. The one arriving second gets a short suffix (for example `VB-2026-11-0005-3DBE`) instead of being rejected, so no sale is lost.

**Q: The website was slow to open in the morning. Why?**

**A:** The backend runs on Render's free plan, which puts the server to sleep after 15 minutes with no visitors; the first request then takes up to a minute. Two pingers keep it awake: UptimeRobot and a GitHub Actions job every 10 minutes.

> **Be honest:** A paid plan removes the sleep entirely; the free plan was chosen to keep hosting at zero cost.

**Q: An update breaks a database migration. Does the whole site go down?**

**A:** No. The server is set to start even if a migration fails; the error is logged loudly instead. Migrations are also written so a half-finished one can safely run again.

## 2. Inventory and medicines

**Q: Two boxes of the same vaccine have different expiry dates. Which is used first?**

**A:** The one that expires soonest. This is FEFO (first expired, first out). Products without an expiry date, like gauze, are used oldest-first. The batch list on screen shows the exact order stock will be used in.

**Q: Could a pet be given an expired vaccine through the system?**

**A:** No. Expired batches are never deducted. If the only stock left has expired, the invoice is refused with a message to restock. Stock expiring today can still be used today.

**Q: A manufacturer recalls lot LOT-123. Can you find which pets received it?**

**A:** The data is there. Every product that tracks expiry must have a lot number, and every deduction records which batch it came from and which invoice (and so which pet) used it.

> **Be honest:** There is no one-click recall screen yet; the trace is done from the records.

**Q: The clinic runs out of a medicine without anyone noticing. How does AutoVet prevent that?**

**A:** Each product has a reorder level. When a sale or service brings total stock across all batches down to that level, a low-stock alert is raised. The AI forecast also predicts the stock-out date, for example "Bordetella vaccine: 0.39 used per day, runs out in 330 days".

**Q: A service like an anti-rabies shot also uses a vaccine. Does stock go down automatically?**

**A:** Yes. Services can be linked to consumables, and finalizing the invoice deducts them along with sold items. If a consumable is short, the invoice still goes through and the admin is notified to reorder.

**Q: How does the AI know when an item will run out? What if it's a new item?**

**A:** It fits a linear regression (scikit-learn) to the item's real usage history: total used over time. The slope is the daily usage rate, and it is checked on held-out data (an 80/20 split, with $R^2$, RMSE and MAE). A new item needs at least 3 usage records on 2 different days; until then it shows "Insufficient Data" instead of guessing.

## 3. Appointments

**Q: Two clients book the same time slot. Can the vet be double-booked?**

**A:** Each time slot accepts at most 2 approved appointments. When an admin approves a booking for a full slot, the system blocks it and asks to decline or move it. Approvals are locked so two admins can't approve into the last place at the same moment.

**Q: A client books the same pet many times in one day. Is that allowed?**

**A:** No. Through the portal, a pet can have at most 2 active bookings per day, and online bookings always start as "pending" until the clinic approves them.

**Q: Can a client book for someone else's pet?**

**A:** No. The portal only allows bookings for the client's own pets; anything else is refused with 403 Forbidden.

## 4. Security and privacy

**Q: A staff member tries to delete a medical record. What happens?**

**A:** It is refused. Only clinical roles (veterinarian, clinic admin) can delete medical records; staff can view them but get "403 Forbidden" on delete. This is enforced on the server, not just hidden in the screen.

**Q: Can a pet owner reach the admin settings or staff list?**

**A:** No. Admin areas check the user's role on every request. A client account gets 403 on settings, staff accounts and the audit trail; tested with Postman for every role.

**Q: Someone tries thousands of passwords on one account. What stops them?**

**A:** Login is limited per email address: after 10 attempts in a minute, more are refused for a while. Sign-up is limited too (per visitor and per email address), and email verification is required before a portal account works.

**Q: Who changed a record, and when?**

**A:** The audit trail records who acted (name and role, kept even if the account is later deleted), what record, what changed, and when. Passwords and tokens are never stored in it.

**Q: Could a stranger send fake data to the cloud sync?**

**A:** No. Every sync request is signed with a secret only the clinic computer and the cloud share, and expires after 5 minutes, so it can't be forged or replayed.

## 5. Billing and records

**Q: How is VAT handled?**

**A:** Prices already include 12% VAT. The system extracts the VAT for the receipt instead of adding it on top: ₱1,120 contains ₱120 VAT. Discounts are applied first, and VAT follows the discounted total.

**Q: A finalized invoice was wrong. Can staff turn it back into a draft?**

**A:** No. A finalized invoice has already deducted stock, so returning it to draft is blocked; that used to let stock disappear with no invoice to show for it. The system asks staff to cancel the invoice instead, so the record of what happened is kept. Paid and cancelled invoices can't be edited.

**Q: A staff member deletes a client by mistake. Is it gone forever?**

**A:** No. Deleted records go to the archive and can be restored for 60 days. Records still linked to other data, like a pet with visits, can't be permanently deleted. A client who deletes their own portal account has a 30-day grace period to change their mind.

**Q: The database is damaged. How do you recover?**

**A:** Admins can create and download a database backup at any time from the system, and a backup is also scheduled daily at 3 AM where the scheduler runs. The downloaded backup is imported back into MySQL to return the data to that point.

> **Be honest:** Restoring is done by importing the backup file; there is no one-click restore button.

## 6. Performance and reliability

**Q: What if many staff use the system at once?**

**A:** We load-tested 30 simultaneous users for 30 seconds: 1,267 requests with 0% errors. The dashboard loaded in 0.16 seconds on average and 95% of loads finished under 0.28 seconds, well under our 3-second target.

> **Be honest:** That test ran locally with production's settings. The live free server has much less CPU, so it is slower there.

**Q: How do you know a new change didn't break something?**

**A:** Automated tests. GitHub runs the backend test suite (PHPUnit, over 100 tests) and a code-style check on every push. We also have frontend tests (Vitest) and Postman collections for the AI module, the full patient journey and role security.

**Q: A user types the wrong password. Does the screen say something misleading?**

**A:** No. The login page tells problems apart: wrong password, blocked or unverified account, too many attempts, server not responding, or no internet, each with its own clear message.